

Rectified Flow Distillation via 2nd Order Inversion for Consistent Text-to-3D Generation

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Abstract

Building on advances in generative diffusion models, recent text-to-3D methods leverage 2D diffusion priors via Score Distillation Sampling (SDS) to optimize 3D content from text prompts. Rectified Flow (RF) has emerged as a deterministic alternative to conventional diffusion and has been integrated into SDS-based pipelines for this task. However, the guidance mechanisms employed in RF-based distillation lack rigorous theoretical justification. In this work, we provide a mathematical analysis showing that commonly used stochastic guidance introduces inconsistent optimization signals, leading to over-smoothing and loss of detail. Motivated by this analysis, we propose a principled deterministic guidance strategy based on high-fidelity second-order ODE inversion. Experiments demonstrate that our method improves 3D consistency and preserves fine geometric details.

Keywords— 3D Generation, Score Distillation Sampling, Rectified Flow, Inversion, ODE Solvers

I. INTRODUCTION

The success of large-scale 2D generative models has sparked interest in 3D generation [27, 9], but creating high-quality 3D assets remains challenging due to limited data and computational costs. Many approaches address this by adapting pretrained 2D text-to-image models [29, 31] to guide 3D optimization [24, 26, 12]. Notably, Score Distillation Sampling (SDS) [27] and Score Jacobian Chaining (SJC) [36] leverage diffusion models as priors, inspiring numerous improvements to the SDS process [39, 4, 7, 10, 15, 22, 38, 42, 43, 44].

Recently, Rectified Flows (RFs) [17, 18] have emerged as a promising alternative to diffusion models, offering stable and direct generation trajectories. RFs approximate a bijective linear mapping between noise and data through straight-line trajectories, employing a simple Ordinary Differential Equation (ODE) to learn a time-independent vector field. Due to these advantages, RF models are adapted

in image [3, 13] and video generation [35], positioning themselves as a compelling method for efficient and high-quality generative tasks. Naturally, this successful paradigm shift has led to the development of RF-based distillation text-to-3D methods that adapt the principles of SDS to the RF-based framework [14, 41, 40].

FlowDreamer [14] and RF-plug-and-play [41] proposed RF-based distillation for text-to-3D generation while addressing inherent problems of SDS-based methods, including mode-seeking behavior and noisy gradients that result in low diversity, blurry, oversaturated, or geometrically inconsistent 3D assets. The fundamental issue stems from the stochastic noise sampling employed at each optimization step, which generates conflicting guidance signals that lead to problematic 3D outputs. Inspired by diffusion model-based SDS approaches [21, 40], RF-based distillation used inversion-based deterministic noise selection to resolve the issue. However, these methods relied on unstable inversion processes and rigorously reformulated mathematical underpinnings for the RF framework were yet limited.

This paper bridges that gap by reformulating RF distillation to correctly incorporate deterministic guidance. Our key insight is that for RFs, inversion quality, sampling quality, and path consistency are all proportionally related. This leads to our key idea: by notably improving the quality of the ODE inversion using a high-fidelity 2nd-order solver [37, 2, 6], as inspired by its success in 2D editing [8, 23, 25], we can find a single, canonical latent noise for each rendered image. This establishes a stable and consistent trajectory for pseudo guidance in 3D scene optimizing process, allowing our method to fundamentally resolve the over-smoothing problem in naive methods. We term our method Rectified Flow Distillation Sampling via 2nd-Order Inversion (**RFDSI**).

In summary, our key contributions are:

- We identify the stochastic guidance mechanism in standard rectified flow distillation as a key cause of over-smoothing and detail loss in text-to-3D generation.
- We propose a novel and effective method, RFDSI, which realizes this deterministic guidance by computing a canonical latent noise for each rendered view

using a high-fidelity 2nd-order ODE inversion.

- We demonstrate through extensive experiments that our approach significantly improves the consistency and detail of generated 3D assets, outperforming standard RF distillation.

II. RELATED WORK

Diffusion and Rectified Flow models. Diffusion models have revolutionized generative modeling, enabling high-quality image synthesis through iterative denoising processes [29, 31]. The primitive diffusion model, DDPM [5], formulated generative modeling as learning to reverse a diffusion process that gradually corrupts data by adding noise. The transition to deterministic Ordinary Differential Equation(ODE)-based formulations, initiated by DDIM [33], enabled reproducible generation and inversion. Advanced ODE solvers [19, 20] further accelerated sampling with higher-order methods, highlighting the importance of solver accuracy. Building on this ODE perspective, recent works have proposed Rectified Flow (RF) models [17, 18, 13, 3], which redefine the generative process as a continuous transformation along straight-line trajectories in latent space. RF models adopt a fully deterministic ODE formulation, which enables direct sampling and inversion [17, 37]. This structured approach has been shown to improve both generation quality and computational efficiency compared to traditional diffusion models.

Score and Flow Distillation. Score Distillation Sampling (SDS) [27] pioneered the use of diffusion models as priors for 3D generation, treating the score function as a guidance signal. SDS operates by adding noise to a rendered image, feeding the noisy image to a denoising model, and comparing the predicted denoising direction with the added noise. This sequence effectively serves as a likelihood function for 3D optimization. Concurrently, Score Jacobian Chaining (SJC) [36] proposed a similar approach which inspired numerous variants that improve gradient quality and stability [39, 4, 7, 10, 15, 22, 38, 42, 43, 33].

Inversion Methods. With the advent of powerful diffusion models, interest in diffusion-based editing tasks naturally increased [11, 25, 8]. These tasks require precise control over the generative process, motivating the use of inversion techniques [25, 34, 23, 8, 6]. Inversion aims to recover the noise latent that reconstructs a given image under a target prompt [32]. The recovered latent captures semantic structure [1, 25] and serves as a foundation for downstream manipulation.

With the emergence of RF models, inversion methods have adapted to exploit their linear trajectory structure [30, 41, 37, 2]. The straight-line formulation simplifies

the underlying ODE, improving compatibility with higher-order solvers. Recent work has shown that solver precision directly impacts inversion quality [6, 37, 2], with 2nd-order solvers providing superior fidelity. This motivates our adoption of a high-fidelity 2nd-order solver for deterministic RF distillation.

III. BACKGROUND

A. Rectified Flow Formulation

Rectified Flow (RF) [17] learns a continuous-time normalizing flow that transforms a noise distribution to a data distribution through straight-line trajectories. Let $x_0 \sim \pi_0$ be a sample from the data distribution and $\varepsilon \sim \pi_1$ be a sample from a Gaussian noise distribution, where $\pi_1 = \mathcal{N}(0, I)$.

Then the RF is defined by an simple ODE:

$$\frac{dx_t}{dt} = v(x_t, t), \quad t \in [0, 1] \quad (1)$$

where $v(x_t, t)$ is the velocity field that defines the flow with data to noise direction and x_t is the state at timestep t . This model transports the data distribution to the noise distribution through ODE.

RF’s forward process connects data and noise in a straight-line trajectory using the following interpolation relationship:

$$x_t = \alpha_t x_0 + \sigma_t \varepsilon \quad (2)$$

where α_t and σ_t are time-dependent coefficients.

The specific formulation can vary. For instance, models like InstaFlow [18] and Flux [13] follow the original RF paper [17], defining the path from noise to data:

$$x_t = tx_0 + (1-t)\varepsilon \quad (\varepsilon \rightarrow x_0) \quad (3)$$

In contrast, Stable Diffusion 3 [3] defines the path from data to noise. In this work, we follow the latter notation:

$$x_t = (1-t)x_0 + t\varepsilon \quad (x_0 \rightarrow \varepsilon) \quad (4)$$

The velocity field can be derived by taking the time derivative of the interpolation path:

$$\dot{x}_t = \frac{dx_t}{dt} = \dot{\alpha}_t x_0 + \dot{\sigma}_t \varepsilon = -x_0 + \varepsilon = \varepsilon - x_0 \quad (5)$$

In optimal conditions, the velocity field predictor $v_\phi(x_t, t)$ is learned to approximate the true velocity $\dot{x}_t$ at any time step t .

B. Rectified Flow Training Objective

Inspired by above observation and equation (5) the training loss for a RF model is:

$$\mathcal{L}_{\text{RF}} = \int_0^1 \mathbb{E}_{\varepsilon \sim \mathcal{N}(0, I)} \left[\|v_\phi(x_t, t) - (\varepsilon - x_0)\|^2 \right] dt \quad (6)$$

where ϕ denotes the learnable parameters of the neural network that parameterizes the velocity field $v_\phi(x_t, t)$.

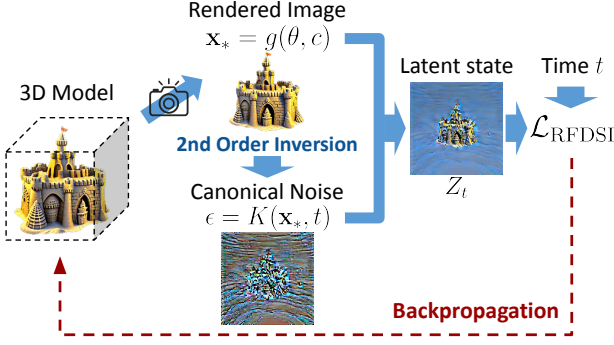


Fig. 1. **RFDSI Pipeline.** 3D model parameters are rendered to an image, inverted to obtain canonical noise, interpolated to form the latent state, and supervised via the RFDSI loss with backpropagation to update the 3D model.

C. Rectified Flow Distillation Sampling (RFDS)

Following the Score Distillation Sampling (SDS) paradigm [27], we adapt the RF training objective for 3D optimization. Given a rendered image $x_* = g(\theta, c)$, where θ denotes the 3D scene parameters and c specifies the camera viewpoint, we derive the RF objective by substituting x_0 with x_* as follows:

$$\mathcal{L}_{\text{RF}} = \int_0^1 \mathbb{E}_{\epsilon \sim \mathcal{N}(0, I)} \left[\|v_\phi(x_t, t) - (\epsilon - x_*)\|^2 \right] dt, \quad (7)$$

where $x_t = \alpha_t x_* + \sigma_t \epsilon$.

We compute the gradient to optimize the 3D scene parameters θ , following [27] and omitting the costly network Jacobian term:

$$\nabla_\theta \mathcal{L}_{\text{RFDS}} = \mathbb{E} \left[w(t) (v_\phi(x_t, t) + x_* - \epsilon) \frac{\partial x_*}{\partial \theta} \right] \quad (8)$$

where $x_t = (1-t)x_* + t\epsilon$ and $w(t)$ is a time-dependent weighting function.

D. Second Order ODE Solver

Studies such as RF-Solve [38] and FireFlow [2] have proposed second-order ODE solvers. In particular, RF-Solve proposes a method to rigorously approximate RF-ODE using Taylor expansion. Since our goal is to analyze and improve through mathematical foundations, we implement a second-order ODE solver using the approach from RF-Solve rather than FireFlow, which is closer to an optimization-based method. The following are the equations for the second-order ODE solver proposed in RF-Solve:

$$dZ_t = v_\phi(Z_t, t) dt \quad (9)$$

$$Z_{t-\tau} = Z_t - \tau v_\phi(Z_t, t) \quad (10)$$

$$Z_{t-\tau} = Z_t - \tau v_\phi(Z_t, t) + \frac{1}{2} \tau^2 v_\phi^{(1)}(Z_t, t) \quad (11)$$

IV. METHOD

A. Limitations: Inconsistent Guidance from Stochastic Paths

The over-smoothing problem arises because different random noise vectors $\epsilon_1, \epsilon_2, \dots$ for the rendered view x^* pull the optimization in conflicting directions. This forces the 3D model parameters θ to converge to a mean representation that satisfies all paths on average, thus losing sharp details and high-frequency information.

B. Inversion Process is solution to Fixed Point Equation (FPE)

From this point onward, we use the notation Z_t to denote the latent variable at time t , replacing the commonly used x_t . We also use ϵ to represent noise in the latent space, with a slight abuse of notation for clarity and consistency. This change reflects our use of a latent-space rectified flow model and is consistent with recent approaches such as [37]. For clarity, we omit the text condition from our equations; however, all processes are text-conditioned, and its inclusion does not affect the overall formulation.

To establish deterministic guidance, we derive the theoretical relationship between RF distillation and the Fixed Point Equation (FPE).

Given a rendered image x_* , we define its latent representation as $Z_* = \Phi(x_*)$, where Φ denotes a image encoder. For a time step t and a pretrained velocity field $v_\phi(Z_t, t)$, the FPE is defined with respect to the latent noise ϵ as:

$$v_\phi(Z_t, t) = \epsilon - Z_*, \quad Z_t = (1-t)Z_* + t\epsilon \quad (12)$$

The concept of inversion is fundamental to understanding the FPE. Inversion methods aim to recover the latent noise ϵ such that the generative trajectory starting from ϵ reaches a given latent point Z_* at time t [32]. This condition is exactly what the FPE encodes: the noise ϵ that satisfies the fixed point equation is the canonical latent that deterministically generates Z_* through the rectified flow. Therefore, any noise obtained through inversion techniques is inherently a solution to the corresponding FPE, establishing the theoretical foundation for our deterministic guidance approach.

We denote the solution ϵ to the FPE as $K(Z_*, t)$, representing the canonical noise that generates Z_* at time t :

$$\epsilon := K(Z_*, t) \quad (13)$$

C. Derivation of RFDSI Loss

Inspired by Score Distillation via Inversion (SDI) [21], we identify a similar distribution-matching behavior in the RF distillation process. Specifically, we note that the difference between consecutive predictions in the reparameterized single-step RF trajectory is analogous to the gradient updates in the SDS formulation.

However, due to the characteristic of Rectified Flow where the single-step prediction converges rapidly to the data distribution, the distributions are not perfectly identical. Despite this, a distinct trend of deblurring in blurry images is observed, enabling it to function as a pseudo-guidance. Based on this observation, we now derive our proposed loss.

The rectified flow forward process at time t is given by:

$$Z_t = (1-t)Z_* + t\varepsilon \quad (14)$$

Then we define the single-step reconstruction estimate from Z_t as $\hat{Z}_0(t)$, following the RF-ODE:

$$\hat{Z}_0(t) = Z_t - t \cdot v_\phi(Z_t, t) \quad (15)$$

$$\hat{Z}_0(t-\tau) = Z_{t-\tau} - (t-\tau) \cdot v_\phi(Z_{t-\tau}, t-\tau) \quad (16)$$

Similarly, the one-step denoising process from the RF-ODE can be written as:

$$Z_{t-\tau} = Z_t - \tau \cdot v_\phi(Z_t, t) \quad (17)$$

The difference between the predictions at time t and $t-\tau$ provides a directional signal for updating the scene parameters. Using equations (15)–(17), we obtain:

$$\begin{aligned} \Delta\hat{Z}_0 &= \hat{Z}_0(t) - \hat{Z}_0(t-\tau) \\ &= (t-\tau) \{v_\phi(Z_{t-\tau}, t-\tau) - v_\phi(Z_t, t)\} \end{aligned} \quad (18)$$

Substituting the expression from Eq. (12) into $\Delta\hat{Z}_0$, we obtain:

$$\Delta\hat{Z}_0 = (t-\tau) \{v_\phi(Z_{t-\tau}, t-\tau) + Z_* - \varepsilon\} \quad (19)$$

This leads to the gradient of the RFDSI loss:

$$\begin{aligned} \nabla_\theta \mathcal{L}_{\text{RFDSI}} &= \mathbb{E}_{\varepsilon, t} \left[w(t) \cdot \left(v_\phi(Z_{t-\tau}, t-\tau) \right. \right. \\ &\quad \left. \left. + Z_* - \varepsilon \right)^\top \frac{\partial Z_*}{\partial \theta} \right] \end{aligned} \quad (20)$$

Here, $w(t)$ is a time-dependent weighting term, and the gradient $\frac{\partial Z_*}{\partial \theta}$ reflects how the latent code changes with respect to the 3D scene parameters θ . By applying the Fixed Point Equation (FPE) in Eq. (12), we substitute the canonical noise solution $K(Z_*, t)$ for ε . Accordingly, the latent state at time $t-\tau$ becomes:

$$Z_{t-\tau} = (1-t+\tau)Z_* + (t-\tau)K(Z_*, t) \quad (21)$$

This yields the final expression for the RFDSI gradient:

$$\boxed{\nabla_\theta \mathcal{L}_{\text{RFDSI}} = \mathbb{E}_t \left[w(t) \cdot \left(v_\phi(Z_{t-\tau}, t-\tau) + Z_* - K(Z_*, t) \right)^\top \frac{\partial Z_*}{\partial \theta} \right]} \quad (22)$$

Comparison with RFDS. The key difference between standard RFDS (Eq. (8)) and RFDSI is to use the deterministic canonical noise $K(Z_*, t)$ instead of random ε , and operates at time $t-\tau$ instead of t .

D. Proposed: Rectified Flow Distillation Sampling via 2nd-Order Inversion (RFDSI)

The core idea of our approach is to compute the canonical latent $K(Z_*, t)$ for a given rendered image x_* , by inverting the rectified flow trajectory through a high-fidelity second-order ODE solver.

2nd-Order Inversion. Instead of relying on a simple first-order Euler approximation, we employ a more accurate 2nd-order ODE solver (e.g., Heun’s method) to solve the reverse ODE and find the canonical noise $K(Z_*, t)$. This approach is inspired by recent successes in 2D image editing [37, 2].

The 2nd-order solver provides significantly better accuracy in solving the fixed point equation compared to first-order methods, ensuring that we obtain a high-quality canonical latent that truly represents the deterministic path for the given image.

Deterministic Flow Distillation Loss. Our final RFDSI loss replaces the stochastic noise ε with the deterministically computed canonical noise $K(Z_*, t)$, enabling consistent optimization across training iterations. This formulation removes the variance introduced by random sampling, allowing the model to follow a stable and semantically meaningful trajectory in latent space. The key insight is that by solving the fixed point equation through high-fidelity 2nd-order inversion, we obtain a canonical noise that provides stable, deterministic guidance for 3D optimization.

V. EXPERIMENTS

A. Setup

Baselines. We compare our proposed RFDSI with state-of-the-art RF-based text-to-3D methods, including FlowDreamer [14], RFDS [41], and RFDS-Rev, a two-stage variant of RFDS. We use the official implementations for RFDS and RFDS-Rev, and reimplement FlowDreamer based on the paper due to the lack of public code.

Implementation Details. For our experiments, we use the `openai/clip-vit-base-patch32` model for CLIP-based prompt alignment evaluation [28], and the `clip-flant5-xl` model for VQA-Score human preference evaluation [16]. The timestep t is sampled from a uniform distribution, initially following $t \sim U[0.02, 0.98]$. After 5000 optimization steps, the sampling range is annealed to $t \sim U[0.02, 0.7]$ to refine the details of the generated assets.

B. Qualitative Results

Fig. 2 compares RFDSI with FlowDreamer [14], RFDS [41], and RFDS-Rev on a range of text prompts. RFDSI consistently produces sharper details and more accurate geometry, mitigating over-smoothing observed in

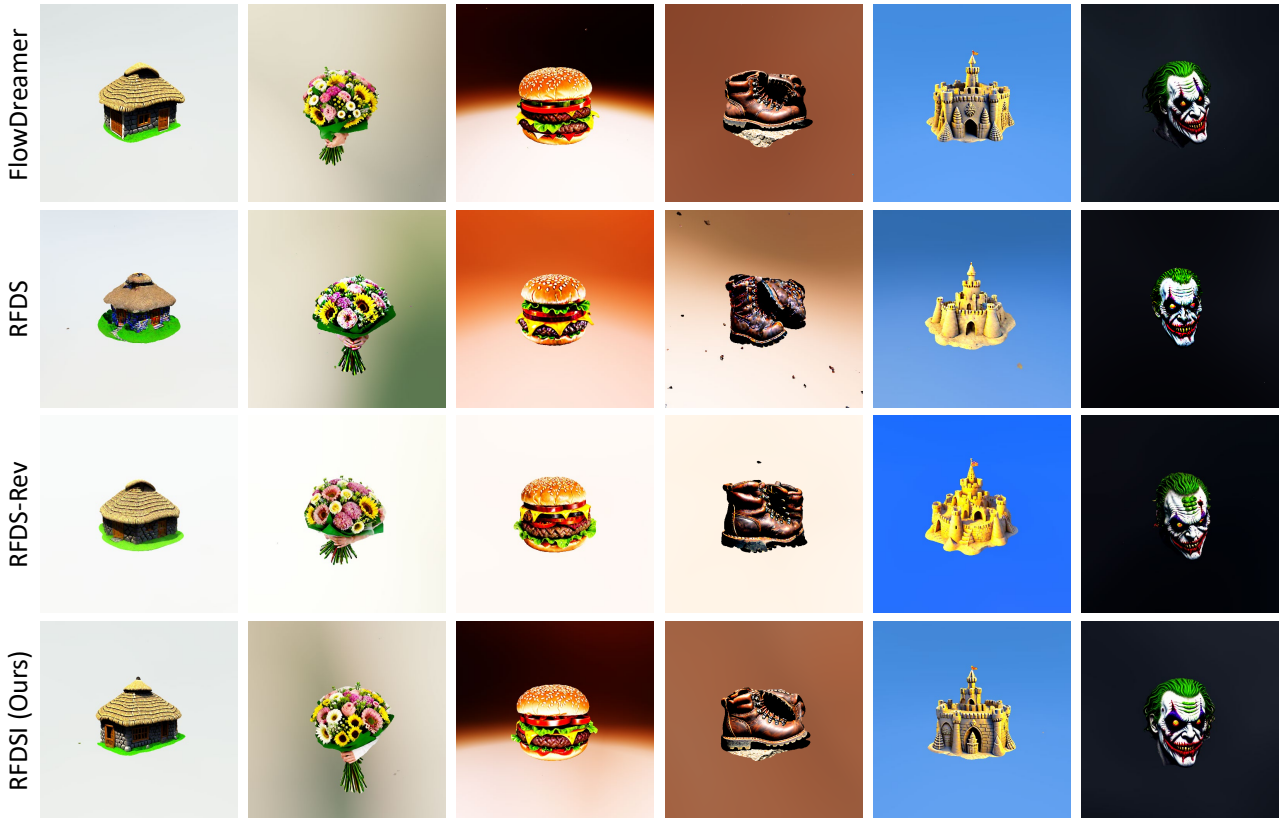


Fig. 2. **Qualitative results.** Our method produces sharper details and better geometric consistency across diverse text prompts. RFDSI effectively mitigates the over-smoothing issues present in standard rectified flow distillation, leading to higher-quality 3D assets with improved fidelity and detail preservation. **Text prompts (left to right):** “A 3D model of an adorable cottage with a thatched roof”, “A florist is making a bouquet with fresh flowers”, “A DSLR photo of a hamburger”, “A DSLR photo of A Rugged, vintage inspired hiking boots with a weathered leather finish, best quality, 4K, HD”, “A highly detailed 3D model of sand castle”, “Zombie JOKER, head, HDR, photorealistic, 8K”.

prior methods. Notably, our method better preserves fine structures (e.g., flowers, hands, textures, facial features) and yields more coherent shapes in complex prompts such as “boots” and “zombie head”. These results highlight the effectiveness of deterministic guidance in improving 3D fidelity and consistency.

C. Quantitative Results

To evaluate the text-image alignment quality, we compute CLIP scores [28] using the OpenAI CLIP-ViT-Base-Patch32 model. For each method, we generate 3D assets from 13 diverse text prompts and render 120 common viewpoints for each generated 3D object. The CLIP scores are computed between the text prompts and the rendered images, then averaged across all views and prompts.

Table 1 shows that our RFDSI method consistently outperforms the baseline RFDS-Rev approach.

The improvement in CLIP score demonstrates that our deterministic guidance mechanism not only enhances visual quality but also improves semantic alignment between the generated 3D assets and the input text prompts. Furthermore, we evaluate our method using the VQA score [16], which serves as a proxy for human preference.

Table 1. Quantitative evaluation using CLIP scores and VQA scores. Results are averaged over 13 text prompts. For CLIP scores, 120 viewpoints per object are used.

Method	CLIP Score ($\uparrow$)	VQA Score ($\uparrow$)
FlowDreamer [14]	25.70	0.72
RFDS [41]	25.28	0.72
RFDS-Rev [41]	24.73	0.69
RFDSI (Ours)	25.71	0.75

As shown in Table 1, our RFDSI achieves the highest VQA score. While the VQA score also measures prompt alignment, it is designed to better reflect human perceptual quality judgments. This result suggests that our method not only improves objective quality and prompt alignment but also generates 3D assets that are more aligned with human preferences compared to the baselines.

VI. CONCLUSION

We address the issue of inconsistent guidance in Rectified Flow-based text-to-3D distillation by introducing RFDSI, a deterministic guidance mechanism based on high-fidelity second-order ODE inversion. Our method es-

establishes a stable optimization trajectory, mitigates over-smoothing, and enables the generation of 3D assets with improved geometric fidelity and detail. While the quality remains bounded by the pretrained 2D Rectified Flow model and the inversion process incurs additional computational cost, our approach demonstrates a substantial improvement in consistency over existing methods. Future directions include exploring higher-order solvers and extending deterministic guidance to broader diffusion-based frameworks.

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SUMMARY OF THIS PAPER

A. Problem Setup

We address the inconsistent guidance problem in Rectified Flow-based 3D distillation. Unlike diffusion models, RF models require specific adaptations for deterministic guidance due to their fundamentally different mathematical formulations.

B. Novelty

Our key innovation is RFDSI (Rectified Flow Distillation Sampling via 2nd-Order Inversion), which computes canonical latent noise vectors using high-fidelity 2nd-order ODE solvers. This establishes deterministic trajectories that eliminate conflicting guidance signals.

C. Algorithms

We reformulate the RF distillation process by replacing stochastic noise sampling with deterministic inversion. For each rendered view, we solve the reverse ODE using a 2nd-order solver to find the unique canonical latent that defines a consistent optimization path.

D. Experiments

Extensive experiments demonstrate that RFDSI significantly outperforms standard RF distillation and other baselines. Our method produces sharper 3D assets with better detail preservation and reduced over-smoothing. Ablation studies confirm the importance of 2nd-order inversion accuracy for establishing consistent guidance.